Nappi: Digital Twin Baby Sleep Monitor

Simulation

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The Problem

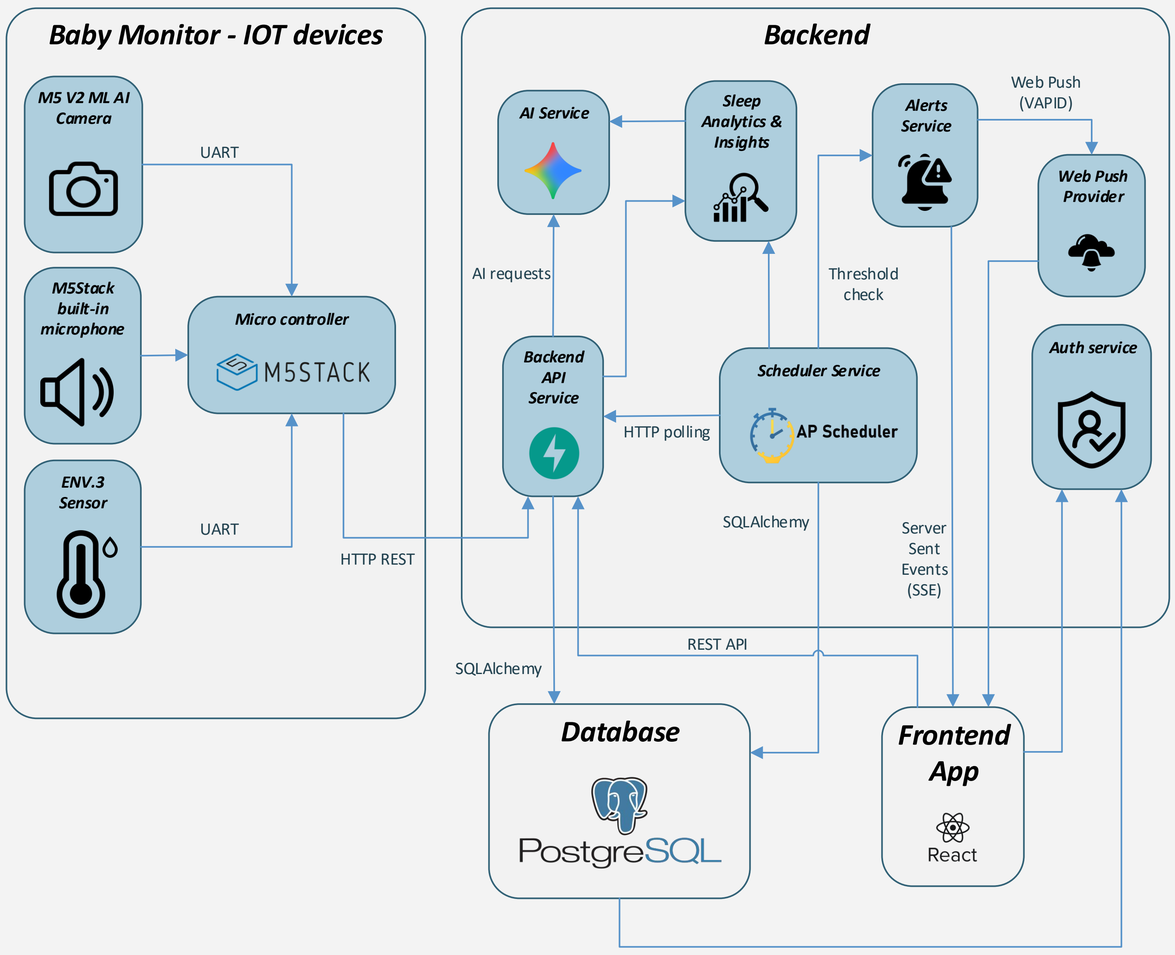
Many parents deal with frequent infant night awakenings, but still don't know why it happened and end up guessing.

- Environmental factors are potential triggers, yet most home device don't track them over time
- Clinical sleep studies are expensive and require in-clinic visits, inaccessible for the average family
- Limited long-term baby data
- Pediatric sleep advice remains generic, unadapted to each baby's individual patterns and environment

Essential Requirements

- **Data Integrity:** Continuously log temperature, humidity, and noise with timestamps, enabling retrospective trend
- **Performance**: Deliver alerts to the dashboard instantly via SSE, with web push fallback when the app is closed
- **Functionality:** Generate awakening insights by analyzing sensor-change magnitude in a configurable window before each wake-up event
- **Usability:** Provide a context-aware AI chat that answers parent questions using the baby's full stored context

System Architecture



Evaluation

Aspect	Requirement	Status
Real-time Monitoring	Sensor data polled every 5s with live SSE streaming	✓
AI Context Accuracy	≥ 80% answers are relevant and baby-specific	90%
Alert Delivery Latency	≤ 2 seconds	✓
Statistics Screen Render Time	≤ 2 seconds per chart	✓

Proposed Solution

A digital-twin inspired platform with:

- **Real-time environment tracking** of temperature, humidity, and noise throughout every sleep session in the baby's room
- **Correlation-based awakening insights** that analyze what changed before a wake-up and suggest likely contributing factors using Google Gemini
- **Smart alerts** delivered instantly via SSE and web push when room conditions deviate from the baby's optimal ranges
- **Interactive AI chat** that answers parent questions using baby's own sleep history, environment data
- **Sleep trend analytics** with pattern detection, historical reporting, and age-based schedule predictions

Key Algorithms

- **Sleep Block Grouping:** Merges closely spaced awakening events into consolidated sleep blocks for cleaner session analysis
- **Awakening Correlation:** Detects which environmental factors changed significantly in the window leading up to a wake-up event
- **Pattern Clustering:** Groups sleep sessions by time-of-day to identify recurring sleep and nap patterns
- **Trend Analysis:** Scores sleep consistency over time to track whether the baby's sleep is improving or declining
- **Schedule Prediction:** Estimates the next likely sleep window using age-appropriate wake window guidelines.
- **Optimal Conditions:** Derives the room conditions most associated with the baby's best sleep sessions

Challenges

- Real-time environment tracking of temperature, humidity, and noise throughout every sleep session in the baby's room
- Real-time delivery complexity across SSE and web push required careful state management across concurrent users
- TinyML model limitations with a small dataset made reliable awake/asleep classification difficult in varied lighting.
- Prompt-engineering Gemini to respond as suggestions, not medical facts while using baby context

Hardware & Enclosure

Nappi IoT hardware includes: M5Stack Core2 microcontroller & built-in SPM1423 digital microphone, M5 Unit V2 ML AI Camera module, and M5 ENV.3 temperature & humidity sensor.